

SONA COLLEGE OF TECHNOLOGY
(AUTONOMOUS INSTITUTION)
DEPARTMENT OF INFORMATION TECHNOLOGY
MINI PROJECT – U19IT703 (2025-26)

DATE: 24.07.2026

BATCH NO:	15	
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TITLE OF THE PROJECT:	UNIVERSAL SYNTHETIC MEDIA FORENSICS ENGINE	
DOMAIN NAME/AREA:	CYBERSECURITY & DIGITAL FORENSICS	
Algorithm/Technique used:	Vision Transformers (ViT), Mel-Spectrogram Extraction, Benford's Law Statistical Analysis,	
REFERENCE (JOURNAL/ ONLINE RESOURCES/ OTHERS)	https://arxiv.org/abs/2010.11929 , https://arxiv.org/abs/2111.02599	
GUIDE NAME & SIGNATURE	MR. A. NAVEEN KUMAR, AP / IT	

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ABSTRACT

The proliferation of advanced Generative AI has outpaced traditional cybersecurity measures, rendering conventional "black-box" deepfake detectors—which focus exclusively on human facial biometrics—obsolete against a new wave of multi-modal synthetic threats, including AI-generated financial documents bypassing KYC protocols, neural voice cloning, fabricated event imagery, and synthetic medical scans. To systematically neutralize these multi-domain vulnerabilities, we propose the Universal Synthetic Media Forensics Engine (USMFE), a patentable edge-to-cloud architecture employing a Dynamic Format-Routing protocol. Upon ingestion via a cross-platform mobile application, the edge device performs real-time live perspective correction and frame extraction, seamlessly routing the payload to five specialized cloud-based forensic pillars. This unified engine synergizes Spatial Forensics using Hugging Face Vision Transformers (ViT) and Error Level Analysis (ELA) for pixel-tampering detection; Biological Forensics via rPPG heartbeat extraction to ensure human liveness; Acoustic Forensics analyzing high-frequency spectral phase discontinuities above 12kHz inherent in neural vocoders; Financial Forensics applying Benford's Law and Optical Character Recognition (OCR) to mathematically validate the statistical distribution of document digits; and Physical Geometry Forensics verifying 3D vanishing point shadow convergence and anatomical symmetry. By aggregating these distinct mathematical metrics through a Weighted Fusion Algorithm, the USMFE transcends simple binary classification to deliver a fully Explainable AI (XAI) solution. The mobile scanner interface provides real-time, pocket-sized authenticity verification by projecting Augmented Reality (AR) bounding boxes and PyTorch Grad-CAM heatmaps directly onto the physical document or screen, accompanied by exact percentage-based confidence scores and human-readable forensic flags, thereby establishing a robust, enterprise-grade framework capable of mathematically dismantling both biometric and non-biometric synthetic media.

PROJECT COORDINATOR